

On the most informative Boolean function under noise

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Abstract. Let X be a uniform random bit string and Y be obtained by copying X and then flipping each of its bits with probability p independently. Let B be a random bit obtained as a function of Y . We show that

$$\mathbf{I}(B : X) \leq \mathbf{I}(Y_1 : X)$$

for all $p \in [0, 1]$, which resolves a conjecture posed by Courtade and Kumar in 2013.

1 Introduction

Let X be distributed uniformly on $\{0, 1\}^n$ and let Y be obtained by copying X and then flipping each of its coordinates Y_i with probability p , independently for $i \in [n]$. In [8], Courtade and Kumar ask, what is the best strategy to output a bit B , having access to Y only, so that $\mathbf{I}(B : X)$, the information revealed about X , is maximized.

They conjecture that taking $B = Y_i$ for some $i \in [n]$ maximizes the information conveyed about X . In particular, their Conjecture 1 states that

$$\mathbf{I}(B : X) \leq \mathbf{I}(Y_1 : X) = 1 - H_2(p). \quad (\text{CK})$$

They further remark that, despite its resemblance to a freebie homework exercise, the statement appears to involve substantial depth. Indeed, despite extensive efforts and partial progress [1, 2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 14, 13, 15, 16, 17, 18], a full resolution of the conjecture remained elusive.

In this work, we show that if we study the inequality parametrized by $\mu = \mathbb{E}[B]$, a delicate refinement of (CK) emerges, which admits an induction on n , the number of coordinates of X, Y . In particular, we prove:

Theorem 1.1. *Let X be a uniform random bit string and Y be obtained by copying X and then keeping each coordinate as-is with probability ρ , replacing it with a uniform random bit with probability $1 - \rho$, independently. Let B be a random bit obtained as a function of Y with expectation μ . Writing $\nu = 4\mu(1 - \mu)$, we have*

$$\mathbf{I}(B : X) \leq \rho^2 (H_2(\mu) - \nu) + \nu \mathbf{I}(Y_1 : X)$$

for any $\rho, \mu \in [0, 1]$.

Two remarks are in order. First, note that an alternative description of Y is to keep each coordinate of X with probability $\rho = 1 - 2p$ and replace it with a uniform random bit with probability $1 - \rho$. Flipping a bit with probability p or re-randomizing with probability $1 - \rho$ leads to the same distribution. Secondly, setting

$$Q_\rho(\mu) = \rho^2 (H_2(\mu) - \nu) + \nu \mathbf{I}(Y_1 : X), \quad (\text{Q})$$

we have $Q_\rho(\mu) \leq \mathbf{I}(Y_1 : X)$ for all $\rho, \mu \in [0, 1]$, whose proof we defer to Section 2.

1.1 The continuous case

We note that $\mathbf{I}(B : X) \leq Q_\rho(\mu)$ is a surprisingly sharp bound, which is already violated by the most natural generalization of the problem. Let $X_\circ$ be distributed uniformly on the unit cube $[0, 1]^n$ and $Y_\circ$ be obtained by copying the corresponding coordinate of $X_\circ$ with

probability ρ and drawing uniformly from $[0, 1]$ with probability $1 - \rho$, for each coordinate $i \in [n]$ independently. Let $B_\circ$ be a random bit obtained as a function of $Y_\circ$ with $\mu = \mathbb{E}[B_\circ]$.

Denote by C_μ the indicator of the event $Y_\circ[1] \leq \mu$, that is, the first coordinate of $Y_\circ$ being at most μ . Note that $\mathbb{E}[C_\mu] = \mu$. In Lemma A.1 we show that $Q_\rho(\mu) < \mathbf{I}(C_\mu : X_\circ)$ for $0 < \mu < 1$, $\mu \neq 1/2$, and $0 < \rho < 1/2$, which means the threshold functions already exceed the bound $Q_\rho(\mu)$. We further conjecture the following.

Conjecture 1.2. *Let $X_\circ$, $Y_\circ$ and $B_\circ$ be as constructed and C_μ be the indicator of the event $Y_\circ[1] \leq \mu$ as above. It holds that*

$$\mathbf{I}(B_\circ : X_\circ) \leq \mathbf{I}(C_\mu : X_\circ)$$

for all $\rho, \mu \in [0, 1]$.

Note that a proof of Conjecture 1.2 directly implies (CK)—though not our Theorem 1.1—since given a strategy to output B from Y revealing $\mathbf{I}(X : B)$ bits of information, we can quantize $Y_\circ$ by rounding each coordinate to 0 or 1 with threshold $1/2$ and apply the strategy to obtain $B_\circ$. We get $\mathbf{I}(B_\circ : X_\circ) = \mathbf{I}(B : X)$.

1.2 Notation

All logarithms are base 2, except $\ln$, which denotes the natural logarithm. We write

$$\begin{aligned} H_2(a) &= a \log \frac{1}{a} + (1-a) \log \frac{1}{1-a}, \\ D_2(a \| b) &= a \log \frac{a}{b} + (1-a) \log \frac{1-a}{1-b} \end{aligned}$$

for the binary entropy and divergence, respectively, with endpoint values understood by continuity. For $a \in [0, 1]$ and $b \in [1/2, 1)$, we repeatedly use

$$\frac{2(a-b)^2}{\ln 2} \leq D_2(a \| b) \leq \frac{\ln(1/b)}{(1-b)^2 \ln 2} (a-b)^2. \quad (\text{U})$$

The identity $D_2(a \| b) = D_2(1-a \| 1-b)$ covers $0 < b < 1/2$ as well. The lower bound also holds for $b = 0, 1$.

For the lower bound, the second derivative of the divergence in a is $1/(a(1-a)\ln 2) \geq 4/\ln 2$, and its value and first derivative vanish at $a = b$. For the upper bound, Taylor's integral formula gives

$$\frac{D_2(a \| b)}{(a-b)^2} = \frac{1}{\ln 2} \int_0^1 \frac{1-s}{(b+s(a-b))(1-b-s(a-b))} ds.$$

The quotient extends continuously to $a = b$ and is convex in a , since $1/(x(1-x)) = 1/x + 1/(1-x)$ is convex on $(0, 1)$. It therefore attains its maximum at an endpoint. The value at $a = 1$ is at least the value at $a = 0$ because

$$-b^2 \ln b + (1-b)^2 \ln(1-b) \geq 0.$$

Indeed, the expression vanishes at $b = 1/2$ and $b = 1$, and its second derivative is $2\ln((1-b)/b) \leq 0$ on this interval. The value at $a = 1$ is the coefficient in the upper bound of (U). Strict convexity of the quotient also makes this bound strict when $0 < a < 1$ and $a \neq b$.

We will also use the Taylor expansion

$$D_2\left(\frac{1+u}{2} \left\| \frac{1}{2}\right.\right) = \frac{1}{2\ln 2} \sum_{k \geq 1} \frac{u^{2k}}{k(2k-1)} \quad (-1 \leq u \leq 1). \quad (1.1)$$

Its value and first derivative vanish at $u = 0$, and its second derivative is $1/((1-u^2)\ln 2)$. Expanding this derivative as a geometric series and integrating twice proves the identity for $|u| < 1$; absolute convergence and continuity include the endpoints.

2 The proof

Our proof proceeds by an induction on n , the number of coordinates of X, Y . Given random variables B and Y , our proof adaptively picks a coordinate $i \in [n]$ and studies

$$\mathbf{I}(B : X) = \mathbf{I}(B : Y_i) - \mathbf{I}(B : Y_i | X) + \mathbf{I}(B : X | Y_i).$$

When Y_i is fixed, B depends on at most $n - 1$ coordinates and therefore the induction hypothesis applies, upper-bounding the last term. The middle two terms, on the other hand, are analyzed in detail in [Section 2.1](#).

Throughout this section, we use the notation $\mu_0 = \mathbb{E}[B | Y_i = 0]$ and $\mu_1 = \mathbb{E}[B | Y_i = 1]$, which satisfy $\mu = (\mu_0 + \mu_1)/2$ since Y_i is distributed uniformly. Recall that $\nu = 4\mu(1 - \mu)$. Similarly, we define $\nu_0 = 4\mu_0(1 - \mu_0)$ and $\nu_1 = 4\mu_1(1 - \mu_1)$ and observe the simple but useful identity

$$\frac{\nu_0 + \nu_1}{2} = \nu - (\mu_1 - \mu_0)^2. \quad (\text{V})$$

As the final step before our proof, let us verify that [Theorem 1.1](#) indeed implies (CK).

Lemma 2.1. *For all $\rho, \mu \in [0, 1]$, we have $Q_\rho(\mu) \leq \mathbf{I}(Y_1 : X)$.*

Proof. Recall that $Q_\rho(\mu) = \rho^2 (\mathbf{H}_2(\mu) - \nu) + \nu \mathbf{I}(Y_1 : X)$. By (U),

$$\mathbf{D}_2(\mu \| 1/2) \geq \frac{1 - \nu}{2 \ln 2}, \quad \mathbf{I}(Y_1 : X) \geq \frac{\rho^2}{2 \ln 2}.$$

Consequently,

$$\begin{aligned} \mathbf{I}(Y_1 : X) - Q_\rho(\mu) &= \rho^2 \mathbf{D}_2(\mu \| 1/2) - (\rho^2 - \mathbf{I}(Y_1 : X))(1 - \nu) \\ &\geq \rho^2 \left(\frac{1}{\ln 2} - 1 \right) (1 - \nu) \geq 0, \end{aligned}$$

noting $\ln 2 < 1$, which completes the proof. $\square$

Define $d_j = 2 |\mathbb{E}(B(2Y_j - 1))|$ for $j \in [n]$. Our induction step always picks a coordinate i with $d_i \leq 1/2$ and the next lemma establishes that such a coordinate exists when $n \geq 2$.

Lemma 2.2. *For any two distinct coordinates i, j , we have $d_i + d_j \leq 1$. In particular, one of d_i, d_j is at most $1/2$.*

Proof. Relabel Y_i, Y_j if necessary so that $\mathbb{E}[B(2Y_i - 1)]$ and $\mathbb{E}[B(2Y_j - 1)]$ are nonnegative. Then

$$d_i + d_j = 2 \mathbb{E}[B(2Y_i - 1) + B(2Y_j - 1)] \leq 4 \Pr[Y_i = Y_j = 1] = 1,$$

since $0 \leq B \leq 1$. $\square$

We fix such an i with $d_i \leq 1/2$ and denote $d = d_i$, which coincides with $|\mu_0 - \mu_1|$ defined above. Next, we would like to bound the middle terms $\mathbf{I}(B : Y_i) - \mathbf{I}(B : Y_i | X)$. However we can control these terms only when mixed with $\frac{1}{2} \mathbf{I}(B : X | Y_i)$, as follows.

Lemma 2.3. *Suppose $d_i \leq 1/2$ and $\mathbf{I}(B : X | Y_i) \leq \frac{1}{2} (Q_\rho(\mu_0) + Q_\rho(\mu_1))$. Then*

$$\mathbf{I}(B : Y_i) - \mathbf{I}(B : Y_i | X) + \frac{1}{2} \mathbf{I}(B : X | Y_i) \leq Q_\rho(\mu) - \frac{1}{4} (Q_\rho(\mu_0) + Q_\rho(\mu_1))$$

holds.

[Section 2.2](#) is dedicated to establishing [Lemma 2.3](#); here we will show how the lemma implies the main theorem.

Theorem 1.1 (restated). *Let X, Y and B be as above. We have $\mathbf{I}(B : X) \leq Q_\rho(\mu)$ for all $\rho, \mu \in [0, 1]$.*

Proof. At $\rho = 0$, both sides are zero. At $\rho = 1$, we have $Y = X$ and both sides equal $H_2(\mu)$. Assume $0 < \rho < 1$. When $n = 1$, the bound holds as B is either a constant, Y_1 or negation of Y_1 . The last two cases hold with equality, whereas the first case reveals zero information.

For $n \geq 2$, Lemma 2.2 applies, giving a coordinate i with $d_i \leq 1/2$. Using the chain rule, we write

$$\mathbf{I}(B : X) = \mathbf{I}(B : Y_i) - \mathbf{I}(B : Y_i | X) + \mathbf{I}(B : X | Y_i).$$

Having fixed Y_i , note that B depends on at most $n-1$ coordinates so our induction hypothesis applies to $Y_i = 0$ and $Y_i = 1$ cases individually, giving

$$\mathbf{I}(B : X | Y_i) \leq \frac{1}{2} (Q_\rho(\mu_0) + Q_\rho(\mu_1)).$$

Summing up the inequality given by Lemma 2.3, we have

$$\begin{aligned} \mathbf{I}(B : X) &= \mathbf{I}(B : Y_i) - \mathbf{I}(B : Y_i | X) + \mathbf{I}(B : X | Y_i) \\ &\leq Q_\rho(\mu) + \frac{1}{2} \mathbf{I}(B : X | Y_i) - \frac{Q_\rho(\mu_0) + Q_\rho(\mu_1)}{4} \\ &\leq Q_\rho(\mu). \end{aligned}$$

completing the proof. $\square$

2.1 The contribution of bit Y_i

Fix the coordinate i chosen above. We first condition on X_{-i} and write

$$a_s = \Pr[B = 1 | X_{-i}, Y_i = s] \quad (s = 0, 1), \quad a = \frac{a_0 + a_1}{2}, \quad t = a_1 - a_0, \quad x = 2a - 1.$$

These are functions of X_{-i} . Since Y_i is uniform and independent of X_{-i} , we have $a = \Pr[B = 1 | X_{-i}]$, $\mathbb{E}[a_s] = \mu_s$ and $|\mathbb{E}[t]| = d$. Also, $|x| + |t| \leq 1$. Set $v = \mathbb{E}[4a(1-a)]$. Then

$$\nu - v = 4 \mathbb{E}[(a - \mu)^2].$$

The first lemma bounds how much information X_i reveals about B , given X_{-i} , in terms of the corresponding information revealed by Y_i .

Lemma 2.4. *Suppose $0 < \rho < 1$ and $d > 0$. Then $d \leq v \leq \nu$ and*

$$v \mathbf{D}_2\left(\frac{1 + d/v}{2} \parallel 1/2\right) \leq \mathbf{I}(B : Y_i | X_{-i}) \leq v. \quad (2.1)$$

Let $\tau \in (0, 1]$ satisfy

$$\mathbf{D}_2\left(\frac{1 + \tau}{2} \parallel 1/2\right) = \frac{\mathbf{I}(B : Y_i | X_{-i})}{v}.$$

Then

$$\mathbf{I}(B : X_i | X_{-i}) \leq v \mathbf{D}_2\left(\frac{1 + \rho\tau}{2} \parallel 1/2\right). \quad (2.2)$$

Proof. Since $|t| \leq 1 - |x| \leq 1 - x^2$, we have $d \leq v$. Also, $\nu - v = 4 \mathbb{E}[(a - \mu)^2] \geq 0$, so $v \leq \nu$. In particular, $v > 0$.

Fix a value of X_{-i} . The information that Y_i reveals about B in this conditional distribution is

$$\ell(x, t) = \frac{1}{2} \left[\mathbf{D}_2\left(\frac{1 + x - t}{2} \parallel \frac{1 + x}{2}\right) + \mathbf{D}_2\left(\frac{1 + x + t}{2} \parallel \frac{1 + x}{2}\right) \right].$$

Conditioned on Y_i , the bit X_i is obtained by flipping Y_i with probability $p = (1 - \rho)/2$, independently of (B, X_{-i}) . Thus

$$\mathbf{I}(B : Y_i | X_{-i}) = \mathbb{E}[\ell(x, t)], \quad \mathbf{I}(B : X_i | X_{-i}) = \mathbb{E}[\ell(x, \rho t)]. \quad (2.3)$$

Computing $\ell(x, t)$ by conditioning on B instead of Y_i , Bayes' rule gives

$$\ell(x, t) = \frac{1+x}{2} \mathbf{D}_2 \left(\frac{1+t/(1+x)}{2} \parallel 1/2 \right) + \frac{1-x}{2} \mathbf{D}_2 \left(\frac{1-t/(1-x)}{2} \parallel 1/2 \right).$$

For now suppose $|x| < 1$, and put $V_x = 1 - x^2$. By (1.1),

$$\frac{\ell(x, t)}{V_x} = \sum_{k \geq 1} \frac{W_k(x)(t/V_x)^{2k}}{2k(2k-1) \ln 2}, \quad W_k(x) = \frac{(1+x)^{2k-1} + (1-x)^{2k-1}}{2}. \quad (2.4)$$

Here $W_1 = 1$ and $W_k \geq 1$. Together with (U), the two expressions for $\ell(x, t)$ give

$$V_x \mathbf{D}_2 \left(\frac{1+t/V_x}{2} \parallel 1/2 \right) \leq \ell(x, t) \leq \frac{t^2}{V_x} \leq V_x, \quad (2.5)$$

where the last inequality uses $|t| \leq 1 - |x| \leq V_x$.

We next compare $\ell(x, \rho t)$ with $\ell(x, t)$. Choose $u \in [0, 1]$ so that

$$\mathbf{D}_2 \left(\frac{1+u}{2} \parallel 1/2 \right) = \frac{\ell(x, t)}{V_x}.$$

Cauchy-Schwarz gives $W_k^2 \leq W_{k-1} W_{k+1}$ for $k \geq 2$. Since $W_1 = 1$ and $W_k \geq 1$, it follows that

$$\ln W_{k+1} - \ln W_k \geq \frac{\ln W_k}{k-1} \quad (k \geq 2),$$

and hence $W_k^{1/k}$ is nondecreasing. The coefficients

$$\frac{W_k(x)(t/V_x)^{2k} - u^{2k}}{2k(2k-1) \ln 2}$$

therefore change sign at most once, from negative to positive. Their sum converges absolutely and is zero by the choice of u . Multiplying them by the decreasing weights ρ^{2k} gives a nonpositive sum: subtract the weight at a sign-change index, and every term is nonpositive. Consequently,

$$\ell(x, \rho t) \leq V_x \mathbf{D}_2 \left(\frac{1+\rho u}{2} \parallel 1/2 \right).$$

This also holds when all the coefficients vanish.

To average this inequality, define F_ρ by

$$F_\rho \left(\mathbf{D}_2 \left(\frac{1+u}{2} \parallel 1/2 \right) \right) = \mathbf{D}_2 \left(\frac{1+\rho u}{2} \parallel 1/2 \right) \quad (0 \leq u \leq 1).$$

We claim that F_ρ is increasing and concave. For $y = \mathbf{D}_2((1+u)/2 \parallel 1/2)$ and $0 < u < 1$,

$$F'_\rho(y) = \rho^2 \frac{\sum_{j \geq 0} \rho^{2j} u^{2j} / (2j+1)}{\sum_{j \geq 0} u^{2j} / (2j+1)} > 0.$$

The ratio decreases with u : differentiating and pairing the terms indexed by j, k gives the sign $(j-k)(\rho^{2j} - \rho^{2k}) \leq 0$. Since y increases with u , the derivative decreases with y , proving concavity. Continuity covers the endpoints.

We can now average with weights V_x/v . Jensen's inequality gives

$$\begin{aligned} \mathbf{I}(B : X_i | X_{-i}) &\leq \mathbb{E} \left[V_x F_\rho \left(\frac{\ell(x, t)}{V_x} \right) \right] \\ &\leq v F_\rho \left(\frac{\mathbf{I}(B : Y_i | X_{-i})}{v} \right) = v \mathbf{D}_2 \left(\frac{1+\rho\tau}{2} \parallel 1/2 \right). \end{aligned}$$

Applying Jensen's inequality to the lower bound in (2.5) similarly gives

$$\begin{aligned} \mathbf{I}(B:Y_i|X_{-i}) &\geq \mathbb{E} \left[V_x \mathbf{D}_2 \left(\frac{1+t/V_x}{2} \parallel 1/2 \right) \right] \\ &\geq v \mathbf{D}_2 \left(\frac{1+\mathbb{E}[t]/v}{2} \parallel 1/2 \right) = v \mathbf{D}_2 \left(\frac{1+d/v}{2} \parallel 1/2 \right), \end{aligned}$$

using symmetry in the last equality. The upper bound in (2.5) gives $\mathbf{I}(B:Y_i|X_{-i}) \leq v$. The lower bound is positive when $d > 0$, so τ is well-defined. When $|x| = 1$, we have $t = 0$ and $\ell(x, t) = 0$; these points contribute zero to the weighted averages. $\square$

The next lemma compares conditioning on X_{-i} with not conditioning on it. Both estimates use the same quantity $\nu - v = 4 \mathbb{E}[(a - \mu)^2]$.

Lemma 2.5. *Suppose $d > 0$, and put*

$$m = |2\mu - 1|, \quad C = \frac{\ln(2/(1+m))}{(1-m)^2}.$$

Then

$$\mathbf{I}(B:Y_i|X_{-i}) - \mathbf{I}(B:Y_i) \geq \frac{d^2(\nu - v)}{2\nu v \ln 2}, \quad (2.6)$$

$$\mathbb{E} \mathbf{D}_2(a \parallel \mu) \leq \frac{C(\nu - v)}{\ln 2}. \quad (2.7)$$

Also, $H(B|X_{-i}) \geq v$ and $\nu C \leq 1$.

Proof. Complement B and relabel Y_i if necessary so that $\mathbb{E}[x] = 2\mu - 1 = m \geq 0$ and $\mathbb{E}[t] = d$. Since Y_i and X_{-i} are independent, the chain rule gives

$$\mathbf{I}(B:Y_i|X_{-i}) - \mathbf{I}(B:Y_i) = \mathbf{I}(Y_i:X_{-i}|B).$$

Conditioned on $(X_{-i}, B = 1)$, the mean of $2Y_i - 1$ is $t/(1+x)$; conditioned on $B = 1$ alone, it is $d/(1+m)$. For $B = 0$, the corresponding means are $-t/(1-x)$ and $-d/(1-m)$. Given X_{-i} , these cases have probabilities $(1+x)/2$ and $(1-x)/2$. Applying the lower bound in (U) and averaging gives

$$\begin{aligned} \mathbf{I}(Y_i:X_{-i}|B) &\geq \frac{1}{2 \ln 2} \left(\mathbb{E} \frac{t^2}{1-x^2} - \frac{d^2}{\nu} \right) \\ &\geq \frac{d^2}{2 \ln 2} \left(\frac{1}{v} - \frac{1}{\nu} \right) = \frac{d^2(\nu - v)}{2\nu v \ln 2}. \end{aligned}$$

The second inequality is Cauchy-Schwarz. At $|x| = 1$, the contribution $t^2/(1-x^2)$ is zero.

For the second bound, $\mu = (1+m)/2 \geq 1/2$, so (U) gives

$$\mathbb{E} \mathbf{D}_2(a \parallel \mu) \leq \frac{\ln(1/\mu)}{(1-\mu)^2 \ln 2} \mathbb{E}[(a - \mu)^2] = \frac{C(\nu - v)}{\ln 2}.$$

Taking $b = 1/2$ in the upper bound of (U) gives $H_2(a) \geq 4a(1-a)$; averaging yields $H(B|X_{-i}) \geq v$. Finally, $\ln(1+u) \leq u$ gives

$$\nu C = \frac{1+m}{1-m} \ln \left(1 + \frac{1-m}{1+m} \right) \leq 1.$$

$\square$

2.2 The conditional mutual information bound

We will combine the two estimates from Section 2.1 to prove Lemma 2.3. We first need the following scalar inequality. Write

$$q = 1 - \rho^2, \quad \beta = \rho^2 - \mathbf{I}(Y_1 : X), \quad \alpha = \frac{\beta \ln 2}{q},$$

and, for $0 < u \leq 1$, set

$$\begin{aligned} \mathcal{A}_\rho(u) &= \mathbf{I}(Y_1 : X) - \mathbf{D}_2((1 + \rho u)/2 \parallel 1/2), \\ \mathcal{R}_\rho(u) &= \rho^2 \mathbf{D}_2((1 + u)/2 \parallel 1/2) - \mathbf{D}_2((1 + \rho u)/2 \parallel 1/2). \end{aligned}$$

Both quantities are nonnegative by (1.1). For $0 < \rho < 1$, we also have $\beta, \alpha > 0$.

A scalar inequality

For the scalar argument, write $r = \rho^2$.

Lemma 2.6 (Scalar inequality). *For $0 < \rho < 1$ and $0 < u \leq 1$,*

$$\frac{\mathcal{A}_\rho(u)}{2(1 + \alpha)} + \frac{2\mathcal{R}_\rho(u)}{u} > \beta. \quad (2.8)$$

We prove the lemma by expanding in r and showing that every coefficient is positive. For integers $N \geq 1$, define the divergence tails

$$b_N = \sum_{k > N} \frac{1}{2k(2k-1)} = \int_0^1 \frac{x^{2N}}{1+x} dx, \quad s_N(u) = \sum_{k > N} \frac{u^{2k}}{2k(2k-1)} = \int_0^u \frac{(u-x)x^{2N}}{1-x^2} dx.$$

Set $b_0 = 1$ as a separate convention, and define

$$\begin{aligned} T_N &= \sum_{i=0}^{N-1} b_i, & B_N &= \sum_{i=0}^{N-1} b_i b_{N-i}, \\ G_N(u) &= \sum_{i=0}^{N-1} b_i s_{N-i}(u), & C_N(u) &= \sum_{k=1}^N \frac{1-u^{2k}}{2k(2k-1)}. \end{aligned}$$

The first ingredient compares the divergence tails in two ways.

Lemma 2.7 (Divergence-tail comparisons). *For $0 < u < 1$ and $N \geq 1$,*

$$G_N(u) \geq B_N \frac{s_N(u)}{b_N}. \quad (2.9)$$

For $N \geq 2$, setting $w = u^{N/2}$ gives

$$\frac{s_N(u)}{ub_N} \geq \frac{s_2(w)}{wb_2}. \quad (2.10)$$

Proof. First, $s_j(u)/b_j$ is the expectation of the decreasing function $(u-x)_+/(1-x)$ under the probability density proportional to $x^{2j}/(1+x)$ on $(0, 1)$. Increasing j tilts this density by an increasing function, so $s_j(u)/b_j$ decreases with j . For $1 \leq j \leq N$, therefore, $s_j(u) \geq b_j s_N(u)/b_N$; summing proves (2.9).

Here is the elementary order principle used in this argument. If Z, Z' are independent with the same distribution, then

$$\text{Cov}(f(Z), g(Z)) = \frac{1}{2} \mathbb{E}[(f(Z) - f(Z'))(g(Z) - g(Z'))].$$

It is nonnegative when both functions increase and nonpositive when one increases and the other decreases. Applying this identity to a density ratio gives the stated comparison of expectations.

For the second comparison, fix $a \geq 0$ and put

$$M_N(a) = \frac{s_N(e^{-a/N})}{e^{-a/N}b_N}.$$

Changing variables in the two tail integrals gives

$$M_N(a) = e^{-2a} \mathbb{E}_{\pi_N} \left[\frac{\sinh(z/N)}{\sinh((z+a)/N)} \right], \quad \pi_N(dz) \propto \frac{e^{-2z}}{1 + e^{z/N}} dz, \quad z > 0. \quad (2.11)$$

The quotient inside the expectation increases with z because $\coth$ decreases. It also increases with N , since $x \coth x$ increases for $x > 0$: its derivative has the sign of $\sinh x \cosh x - x > 0$. Moreover, for $K > N$ the density ratio π_K/π_N increases with z , as its logarithmic derivative is

$$\frac{N^{-1}}{1 + e^{-z/N}} - \frac{K^{-1}}{1 + e^{-z/K}} > 0.$$

The order principle and then the pointwise comparison of quotients show that $M_N(a)$ increases with N . Taking $a = -N \ln u$ gives $e^{-a/N} = u$ and $e^{-a/2} = u^{N/2} = w$, proving (2.10). All the densities are integrable; at $a = 0$ the quotient in (2.11) equals one. $\square$

The second ingredient gives estimates uniform in the coefficient index N .

Lemma 2.8 (Uniform moment bounds). *Let*

$$e_N = \frac{1}{2} \sum_{k=1}^N \frac{1}{2k-1}, \quad R_N = 2T_N - 1, \quad x = 1 - u^{2N}.$$

For $N \geq 2$ and $0 \leq u \leq 1$,

$$B_N \leq b_N R_N, \quad \frac{R_N}{2e_N} < \frac{19}{18}, \quad \frac{C_N(u)}{2e_N b_N} \geq 2x + \frac{3}{4}x^2 + \frac{1}{8}x^3. \quad (2.12)$$

Proof. The tail integral and finite summation give

$$\frac{1}{4N+2} \leq b_N \leq \frac{1}{4N+1}, \quad T_N = 1 - \ln 2 + e_N + N b_N. \quad (2.13)$$

Indeed, the integral bounds follow from $1+x \leq 2$ and $1+x \geq 2\sqrt{x}$. For positive integers i, j , they imply

$$b_i^{-1} + b_j^{-1} \geq 4(i+j) + 2 \geq b_{i+j}^{-1}, \quad b_i b_j \leq b_{i+j}(b_i + b_j).$$

Summing over $i+j = N$ and including the $b_0 b_N = b_N$ term proves $B_N \leq b_N R_N$.

For the second bound, put $t = \sqrt{2/N} \in (0, 1]$. We will use

$$e_N \geq \frac{7}{6} - \frac{t}{2} \geq 1 - \frac{t}{3}. \quad (2.14)$$

The first inequality is equality at $N = 2$. Its right-hand side increases from N to $N+1$ by at most $e_{N+1} - e_N = 1/(4N+2)$, because

$$\sqrt{N}\sqrt{N+1}(\sqrt{N} + \sqrt{N+1}) \geq (2N+1)\sqrt{N} \geq (2N+1)\sqrt{2}.$$

This proves the first inequality by induction; the second uses $t \leq 1$. We also have

$$\frac{9}{13} < \ln 2 < \frac{25}{36}. \quad (2.15)$$

For the lower bound, the first three terms of $\ln 2 = 2 \sum_{j \geq 0} ((2j+1)3^{2j+1})^{-1}$ sum to $842/1215 > 9/13$. For the upper bound, replace $2j+1$ by 3 for every $j \geq 1$ and sum the resulting geometric series. Writing $\gamma = \ln 2 - 1/2 > 5/26$, we obtain

$$\begin{aligned} N b_N &\leq \frac{1}{4 + t^2/2} \leq \frac{1}{4} - \frac{t^2}{36}, \\ 18N b_N - e_N &\leq \frac{10}{3} + \frac{t - t^2}{2} \leq \frac{83}{24} < \frac{45}{13} < 18\gamma. \end{aligned}$$

Since $R_N = 2e_N + 2Nb_N - 2\gamma$, this is equivalent to $R_N/(2e_N) < 19/18$.

For the third bound, apply the first three nonnegative terms of the binomial expansion of $1 - (1 - x)^a$, where $0 < a \leq 1$. Put $s = 1/N = t^2/2$ and $e = e_N$. The moment identities

$$\sum_{k=1}^N \frac{k}{2(2k-1)} = \frac{N}{4} + \frac{e_N}{2}, \quad \sum_{k=1}^N \frac{k^2}{2(2k-1)} = \frac{N(N+2)}{8} + \frac{e_N}{4}$$

follow by dividing the rational summands. Together with $b_N \leq 1/(4N+1)$, they yield

$$\begin{aligned} \frac{C_N(u)}{2e_N b_N} &\geq \frac{4+s}{2} [x + \theta_2 x^2 + \theta_3 x^3], \\ \theta_2 &= \frac{1}{2} - \frac{s}{4} - \frac{1}{8e}, \quad \theta_3 = \frac{1}{3} - \frac{s}{4} + \frac{s^2}{24} - \frac{5-2s}{48e}. \end{aligned} \tag{2.16}$$

By (2.14),

$$e^{-1} \leq (1 - t/3)^{-1} \leq 1 + t/2.$$

Let $A = (4+s)\theta_2/2 - 3/4$ and $D = (4+s)\theta_3/2 - 1/8$. Substitution gives

$$\begin{aligned} A &\geq -\frac{t}{8} - \frac{5t^2}{32} - \frac{t^3}{64} - \frac{t^4}{32} \geq -\frac{21t}{64} \geq -\frac{t}{3}, \\ D - \left(\frac{1}{3} - \frac{t}{4}\right) &\geq \frac{t}{384} [2(1-t)(28-t) + t^2((1-t)^2 + t^3)] \geq 0. \end{aligned}$$

Consequently, the difference between the right-hand side of (2.16) and $2x + 3x^2/4 + x^3/8$ is bounded below by

$$\begin{aligned} x \left[\frac{t^2}{4} - \frac{tx}{3} + \left(\frac{1}{3} - \frac{t}{4}\right) x^2 \right] &= \frac{x}{12} [3(t-x)^2 + 2tx(1-x) + x^2(1-t)] \\ &\geq 0. \end{aligned}$$

This proves the last bound. □

The preceding estimates will reduce every coefficient of index at least two to the following fixed function.

Lemma 2.9 (A fixed divergence inequality). *For $0 < w < 1$,*

$$\mathcal{J}(w) = \frac{55}{72} - \frac{31w^4}{8} + \frac{9w^8}{8} - \frac{w^{12}}{8} + 38 \frac{s_2(w)}{w} > 0. \tag{2.17}$$

Proof. Write $g(w) = s_2(w)/w$. Differentiation gives

$$\begin{aligned} g'(w) &= -\frac{\ln(1-w^2)}{2w^2} - \frac{1}{2} - \frac{w^2}{4} = \sum_{k \geq 3} \frac{w^{2k-2}}{2k}, \\ g''(w) &= \frac{1}{w(1-w^2)} + \frac{\ln(1-w^2)}{w^3} - \frac{w}{2}. \end{aligned}$$

Thus

$$\frac{\mathcal{J}'(w)}{w^3} = -\frac{31}{2} + 9w^4 - \frac{3}{2}w^8 + 19 \sum_{k \geq 3} \frac{w^{2k-5}}{k}$$

is strictly increasing on $(0, 1)$. Also $g'''(w) > 0$, and for $w \geq 4/5$,

$$\left(-\frac{31w^4}{8} + \frac{9w^8}{8} - \frac{w^{12}}{8} \right)''' = w(-93 + 378w^4 - 165w^8) > 0.$$

Indeed, the quadratic $-93 + 378z - 165z^2$ increases on $[0, 1]$ and is positive at $z = 2/5 < (4/5)^4$. Hence $\mathcal{J}'''(w) > 0$ for $w \geq 4/5$.

We need four finite estimates:

$$\mathcal{J}'(4/5) < 0, \quad \mathcal{J}''(4/5) > 25, \quad \mathcal{J}(5/6) > \frac{1}{256}, \quad |\mathcal{J}'(5/6)| < \frac{1}{5}. \quad (2.18)$$

For completeness, the following rational bounds certify them. For $z > 1$, put $v = (z-1)/(z+1)$ and define

$$P(z) = 2 \sum_{j=0}^5 \frac{v^{2j+1}}{2j+1}, \quad E(z) = \frac{2v^{13}}{13(1-v^2)}.$$

The logarithm series gives $P(z) < \ln z < P(z) + E(z)$. Only $z \in \{2, 3/2, 5/4, 11/8\}$ are needed, using

$$\begin{aligned} \ln(9/25) &= 2 \ln(3/2) - 2 \ln(5/4) - 2 \ln 2, \\ \ln(11/36) &= \ln(11/8) - 2 \ln(3/2) - \ln 2, \\ (\ln 2) \mathbf{D}_2(11/12 \| 1/2) &= \frac{3}{4} \ln 2 + \frac{11}{12} \ln(11/8) - \ln(3/2). \end{aligned}$$

Substituting the appropriate lower or upper bound for each logarithm in the displayed formulas for g', g'' , and

$$g(w) = \frac{(\ln 2) \mathbf{D}_2((1+w)/2 \| 1/2) - w^2/2 - w^4/12}{w},$$

gives

quantity	strict lower bound	strict upper bound
$\mathcal{J}'(4/5)$	-1	-9/10
$\mathcal{J}''(4/5)$	25	26
$\mathcal{J}(5/6)$	9/2000	1/200
$\mathcal{J}'(5/6)$	9/50	19/100

These are finite rational comparisons and imply (2.18).

Since $\mathcal{J}'''$ is positive on $[4/5, 1)$, we have $\mathcal{J}'' > 25$ there. Taylor's formula at 5/6 therefore gives

$$\begin{aligned} \mathcal{J}(w) &\geq \mathcal{J}(5/6) + \mathcal{J}'(5/6)(w - 5/6) + \frac{25}{2}(w - 5/6)^2 \\ &> \frac{1}{256} - \frac{1}{1250} > 0 \quad (4/5 \leq w < 1). \end{aligned}$$

On $(0, 4/5]$, the increasing ratio $\mathcal{J}'(w)/w^3$ and the inequality $\mathcal{J}'(4/5) < 0$ show that $\mathcal{J}$ decreases. Its value there is consequently at least $\mathcal{J}(4/5) > 0$. $\square$

Lemma 2.10 (Positive coefficients). *For every integer $N \geq 1$ and $0 < u \leq 1$,*

$$c_N(u) := C_N(u) + \frac{4G_N(u)}{u} - 2B_N > 0.$$

Proof. First suppose $N \geq 2$ and $0 < u < 1$, and put

$$k = \frac{4s_N(u)}{ub_N} - 2, \quad w = u^{N/2}, \quad x = 1 - w^4.$$

By Lemma 2.7, $c_N(u) \geq C_N(u) + B_N k$. If $k \geq 0$, this is positive because $C_N(u) > 0$. If $k < 0$, Lemma 2.8 gives

$$\begin{aligned} \frac{c_N(u)}{2e_N b_N} &\geq \frac{C_N(u)}{2e_N b_N} + \frac{R_N}{2e_N} k \\ &\geq 2x + \frac{3}{4}x^2 + \frac{1}{8}x^3 + \frac{19}{18}k. \end{aligned}$$

Here the upper bounds on B_N and $R_N/(2e_N)$ give lower bounds after multiplication by the negative number k . Since $b_2 = \ln 2 - 7/12 < 1/9$, the rescaled tail comparison gives

$$k \geq 36 \frac{s_2(w)}{w} - 2.$$

Substituting this and $x = 1 - w^4$ into the preceding lower bound gives exactly $\mathcal{J}(w) > 0$. For $u = 1$, we instead have $s_j(1) = b_j$ and $C_N(1) = 0$, so $c_N(1) = 2B_N > 0$.

It remains to treat $N = 1$. Writing $\gamma = \ln 2 - 1/2 < 7/36$ and retaining the first two terms of $s_1(u)$ gives

$$\begin{aligned} c_1(u) &= \frac{1-u^2}{2} + \frac{4s_1(u)}{u} - 2\gamma \\ &\geq Z(u) := \frac{1}{9} - \frac{u^2}{2} + \frac{u^3}{3} + \frac{2u^5}{15}. \end{aligned}$$

To see that $Z > 0$, note that $Z'(u)/u = -1 + u + 2u^3/3$ increases and is negative at $u = 2/3$. Also $Z''(u) = -1 + 2u + 8u^3/3 > 1$ on $[2/3, 1]$. At $a = 3/4$,

$$Z(a) = \frac{49}{23040} > \frac{1}{512}, \quad Z'(a) = \frac{3}{128} < \frac{1}{32}.$$

Taylor's formula on $[2/3, 1]$ yields

$$Z(u) \geq Z(a) + Z'(a)(u-a) + \frac{1}{2}(u-a)^2 > \frac{1}{512} - \frac{1}{2048} > 0.$$

On $(0, 2/3]$, the monotonicity of $Z'(u)/u$ makes Z decreasing, so $Z(u) \geq Z(2/3) > 0$ there as well. $\square$

Proof of Lemma 2.6. The divergence-tail definitions give the absolutely convergent expansions

$$\alpha = \sum_{N \geq 1} b_N r^N, \quad \frac{(\ln 2) \mathcal{R}_\rho(u)}{q} = \sum_{N \geq 1} s_N(u) r^N, \quad \frac{(\ln 2) \mathcal{A}_\rho(u)}{q} = \sum_{N \geq 1} c_N(u) r^N.$$

In particular, $\alpha > 0$. Multiplying the series, and using the convention $b_0 = 1$, gives

$$\frac{\ln 2}{q} \left[\mathcal{A}_\rho(u) + \frac{4(1+\alpha) \mathcal{R}_\rho(u)}{u} - 2\beta(1+\alpha) \right] = \sum_{N \geq 1} c_N(u) r^N.$$

Every coefficient on the right is positive by Lemma 2.10. Since $0 < r < 1$ and $q > 0$, the expression in brackets is positive. Dividing it by $2(1+\alpha)$ proves (2.8). $\square$

Combining the posterior estimates

Proof of Lemma 2.3. At $\rho = 0, 1$, the claimed inequality is an equality, so assume $0 < \rho < 1$. If $d = 0$, then $\mu_0 = \mu_1 = \mu$ and $\mathbf{I}(B : Y_i) = 0$. The claim follows from ?? and $\mathbf{I}(B : Y_i | X) \geq 0$.

Assume $d > 0$. Use v, τ from Lemma 2.4 and m, C from Lemma 2.5. For this proof, write

$$\begin{aligned} h &= \mathbf{H}(B | X_{-i}), & J &= \mathbf{H}(B | X, Y_i), \\ M &= \mathbf{H}(B | X), & J_* &= \mathbf{H}(B | Y_i), \\ K &= \mathbf{I}(B : Y_i | X_{-i}), & K_* &= \mathbf{I}(B : Y_i). \end{aligned}$$

Since X_i is independent of (B, X_{-i}) given Y_i , the chain rule gives

$$K = h - J, \quad K_* = \mathbf{H}_2(\mu) - J_*, \quad M - J = \mathbf{I}(B : Y_i | X).$$

Also, $\mathbf{H}_2(\mu) - h = \mathbb{E} \mathbf{D}_2(a \| \mu)$. Put $z = \nu - v$, $s = d^2$ and

$$j = qJ_* + \beta(\nu - s).$$

Since $J_* = H(B | Y_i)$ and $J = H(B | X, Y_i)$, the assumption ?? is exactly $J \geq j$. We obtain two lower bounds on $\mathbf{I}(B : Y_i | X) = M - J$.

First, Lemma 2.4 and the defining equation for τ give

$$M \geq J + qK + v\mathcal{R}_\rho(\tau).$$

Using Lemma 2.5, we obtain

$$\mathbf{I}(B : Y_i | X) \geq qK_* + \frac{qs}{2\nu v \ln 2} z + v\mathcal{R}_\rho(\tau). \quad (2.19)$$

Second, write the same transport estimate as

$$\begin{aligned} \mathbf{I}(B : Y_i | X) &\geq h - v\mathbf{D}_2((1 + \rho\tau)/2 \| 1/2) - J \\ &= qK_* + \beta s + (j - J) + v\mathcal{A}_\rho(\tau) \\ &\quad + \rho^2(h - v) - q\mathbb{E}\mathbf{D}_2(a \| \mu) - \beta z \\ &\geq qK_* + \beta s + (j - J) + v\mathcal{A}_\rho(\tau) - \frac{q}{\ln 2}(C + \alpha)z. \end{aligned} \quad (2.20)$$

The equality uses $K_* = H_2(\mu) - J_*$, $z = \nu - v$, $\beta = \rho^2 - \mathbf{I}(Y_1 : X)$, and $\mathbb{E}\mathbf{D}_2(a \| \mu) = H_2(\mu) - h$. The last step uses $h \geq v$ and $\mathbb{E}\mathbf{D}_2(a \| \mu) \leq Cz / \ln 2$ from Lemma 2.5.

Multiply (2.20) by

$$\lambda = \frac{s}{2\nu v(C + \alpha)} > 0$$

and add (2.19). The terms involving z cancel, leaving

$$\begin{aligned} (1 + \lambda)\mathbf{I}(B : Y_i | X) &\geq (1 + \lambda)(qK_* + \beta s) + \lambda(j - J) \\ &\quad + s \left[\frac{v}{s}\mathcal{R}_\rho(\tau) + \frac{\mathcal{A}_\rho(\tau)}{2\nu(C + \alpha)} - \beta \right]. \end{aligned} \quad (2.21)$$

By Lemma 2.4, $d/v \leq \tau$. Together with $d \leq 1/2$, this gives $s/v \leq \tau/2$. Also $\nu C \leq 1$ and $\nu \leq 1$, so $\nu(C + \alpha) \leq 1 + \alpha$. The expression in brackets is therefore at least

$$\frac{2\mathcal{R}_\rho(\tau)}{\tau} + \frac{\mathcal{A}_\rho(\tau)}{2(1 + \alpha)} - \beta > 0,$$

where the strict inequality is Lemma 2.6. Consequently,

$$\mathbf{I}(B : Y_i | X) > qK_* + \beta s + \frac{\lambda}{1 + \lambda}(j - J). \quad (2.22)$$

It remains to bound the coefficient of $j - J$. For $b = (1 - m)/(1 + m) \in (0, 1]$, concavity of the logarithm gives

$$\nu C = \frac{\ln(1 + b)}{b} \geq \ln 2.$$

Since $v \geq d$ and $d \leq 1/2$, we have

$$\lambda \leq \frac{d}{2\nu C} \leq \frac{1}{4\ln 2} < 1, \quad \frac{\lambda}{1 + \lambda} < \frac{1}{2}.$$

Using $j - J \leq 0$ in (2.22) now yields

$$\mathbf{I}(B : Y_i | X) > qK_* + \beta d^2 + \frac{1}{2}(j - J).$$

Finally,

$$j - J = \mathbf{I}(B : X | Y_i) - \frac{Q_\rho(\mu_0) + Q_\rho(\mu_1)}{2}.$$

By (V), we also have

$$\frac{Q_\rho(\mu_0) + Q_\rho(\mu_1)}{2} = Q_\rho(\mu) - \rho^2 \mathbf{I}(B : Y_i) + (\rho^2 - \mathbf{I}(Y_1 : X))d^2. \quad (2.23)$$

Substituting these two identities gives

$$\mathbf{I}(B : Y_i) - \mathbf{I}(B : Y_i | X) + \frac{1}{2}\mathbf{I}(B : X | Y_i) < Q_\rho(\mu) - \frac{Q_\rho(\mu_0) + Q_\rho(\mu_1)}{4},$$

as required. $\square$

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A The continuous case

Lemma A.1. *In the continuous problem, let C_μ be the indicator of $Y_\circ[1] \leq \mu$. Then*

$$Q_\rho(\mu) \leq \mathbf{I}(C_\mu : X_\circ)$$

for $\rho, \mu \in [0, 1]$. For $0 < \rho < 1$, equality holds exactly when $\mu \in \{0, 1/2, 1\}$. At $\rho = 0, 1$, equality holds for every μ .

Proof. The cases $\mu = 0, 1/2, 1$ follow directly, so assume $0 < \mu < 1$ and $\mu \neq 1/2$. Conditioned on $X_\circ$, the probability that $C_\mu = 1$ is $\rho + (1 - \rho)\mu$ if $X_{\circ,1} \leq \mu$, and $(1 - \rho)\mu$ otherwise. Hence

$$\mathbf{I}(C_\mu : X_\circ) = \mu \mathbf{D}_2(\rho + (1 - \rho)\mu \parallel \mu) + (1 - \mu) \mathbf{D}_2((1 - \rho)\mu \parallel \mu).$$

Keep μ fixed and write

$$F(\rho) = \mathbf{I}(C_\mu : X_\circ) - \nu \mathbf{D}_2((1 - \rho)/2 \parallel 1/2).$$

We have $F(0) = F'(0) = 0$ and $F(1) = \mathbf{H}_2(\mu) - \nu$. Differentiating the displayed divergence formula twice gives

$$F''(\rho) = \frac{\nu(1 - \nu)(1 - \rho)}{\ln 2 (1 + \rho)(4\rho + \nu(1 - \rho)^2)}.$$

This is strictly decreasing on $[0, 1]$: its numerator decreases, while both factors in its denominator increase. Thus, for $0 < \rho < 1$,

$$\frac{F(\rho)}{\rho^2} = \int_0^1 (1 - t) F''(\rho t) dt > \int_0^1 (1 - t) F''(t) dt = F(1) = \mathbf{H}_2(\mu) - \nu.$$

Adding $\nu \mathbf{D}_2((1 - \rho)/2 \parallel 1/2)$ proves the strict inequality. The cases $\rho = 0, 1$ follow from the values of F at the endpoints. $\square$

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